Communal Computing and Constitutional Governance

Technical Brief on Dynamicland, Public Space, and Governed Shared Reality

The central contribution is architectural rather than empirical.

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Abstract

This technical brief asks how communal, embodied, locally modifiable computing could become a genuine computational public space rather than another private platform presented on a shared surface. Drawing on an author-supplied but presently unidentified Dynamicland transcript, it proposes a synthesis in which visible artifacts, shared physical context, local authorship, privacy by non-collection, governed admission, receipt waymarks, and portable constitutional representations support civic computation.

The central contribution is architectural rather than empirical. Co-location is not common knowledge; visibility is not intelligibility; local control is not legitimate authority; physical possession is not automatically secure custody; and a receipt can become surveillance if it captures more than reconstruction requires. The brief therefore adds explicit primitives, a transition contract, universal-GSM/GMB placement, the complete realization funnel, a bounded pilot configuration, countermodels, failure modes, and publication decisions. It does not claim that a Dynamicland system, UGK integration, MCIR lowering path, civic deployment, or constitutional public space has been implemented or validated.

Keywords: communal computing; constitutional computing; computational public space; Dynamicland; civic infrastructure; local-first systems; privacy by non-collection; receipts; governed admission; public artifacts; GMB; MCIR.

Source and Claim Standing

The exact source draft remains unchanged. Its descriptions of Dynamicland and the quoted transcript claims are treated here as the author's interpretations of a transcript whose exact artifact, speaker, publication metadata, date, URL, and license are not identified in the paper or recoverable through the current Ableman

corpus skill. They require source recovery and primary-source verification for a future evidence-qualified revision. No external research was performed in this pass.

The paper's constitutional synthesis, design principles, and proposed system relations are author material. The definitions, equations, GMB/GSM projection, full realization funnel, pilot configuration, counterevidence program, and decision surface added here are AI-assisted editorial or synthetic contributions explicitly distinguished from the original author material. Source-specific external systems excluded by the author's scope ruling remain visible only in exact source custody and are not integrated into this candidate.

1. Core Thesis

Dynamicland proposes a form of computation that is communal, physical, locally made, publicly inspectable, and embedded in shared reality. Its critique of screen-based computing is not merely ergonomic. It is political, civic, epistemic, and architectural.

The central claim is that contemporary computing isolates people into private, vendor-made worlds. Dynamicland attempts to reverse this by making computation something people can gather around, touch, inspect, modify, teach, and remake together.

This is highly relevant to constitutional governance.

The constitutional governance version of Dynamicland's thesis is:

> Computation should not merely produce outputs. It should create shared, inspectable, governable reality.

Dynamicland contributes the civic and embodied side of the proposal. UGK, MCIR, GMB, and a domain-governed admission profile are candidate constitutional interfaces for making that shared reality receipted, privacy-bounded, locally adaptable, and accountable across time. None is implemented or validated by this paper.

1.1 Operative primitives and non-equivalences

Primitive	Working meaning in this paper
communal computation	computation whose relevant artifacts and actions can be jointly encountered, interpreted, and modified by participants in a shared practice

Primitive	Working meaning in this paper
computational public space	a governed environment in which participation, rules, authority, evidence, changes, dissent, and exit conditions are inspectable and not wholly controlled by an unaccountable private operator
public artifact	an artifact intentionally exposed within a declared public, civic, or communal jurisdiction; public does not mean unrestricted or context-free access
constitutional surface	the human- and machine-readable projection of active rules, authority, state, evidence, refusals, and amendment paths governing an activity
local profile	a community-specific vocabulary, authority structure, workflow, and evidence policy bound to a jurisdiction and portable representation contract
governed admission	the decision process that distinguishes a proposal from an authorized state change
receipt waymark	a deliberately minimal record sufficient to reconstruct a governed transition without capturing the entire surrounding activity
privacy by non-collection	satisfaction of a governance objective while declining to sense, transmit, retain, or derive data not required for that objective
dissent artifact	an objection, minority report, refusal, alternative, or unresolved issue retained without being misrepresented as the active rule
portable core	the smallest canonical representation and transition contract required for exchange without erasing legitimate local variation

The architecture depends on several non-equivalences:

shared surface
$\neq$ shared interpretation, visibility,
$\neq$ intelligibility

proposal right
$\neq$ admission authority, physical possession,
$\neq$ verified custody

local control
$\neq$ legitimate representation, receipt.
$\neq$ complete surveillance log

1.2 Minimal governed-transition contract

Let S_n be the publicly relevant governed state, p a proposal, J the active jurisdiction, A the applicable authority relation, L the declared law or local profile, E the admissible evidence, and B the privacy and custody boundary. A protected transition may produce S_{n+1} only when:

$$\begin{aligned}&\mathrm{Admit}(p, S_n, J, A, L, E, B) = \mathrm{InScope}(p, J)\\ &\wedge \mathrm{Authorized}(p, A)\\ &\wedge \mathrm{Satisfied}(p, L, E)\\ &\wedge \mathrm{PrivacyBounded}(p, B)\end{aligned}.$$

If admitted, the transition produces a scoped receipt waymark:

$$r_n = H(\mathrm{id}(S_n), \mathrm{id}(p), J, A, L, E, B, \mathrm{id}(S_{n+1}), o),$$

where o records the outcome and H denotes a declared commitment function. This is a design contract, not a claim that one universal hash, authority scheme, or receipt format is sufficient. A real profile must also specify refusal, defer, indeterminate effects, appeal, amendment, minority report, and recovery.

1.3 Constitutional invariants proposed by the brief

1. **No hidden law for protected public effects.** Participants need a usable projection of the rule and authority path that governs an effect.
2. **Proposal is not admission.** Broad participation rights need not grant universal mutation authority.
3. **Visibility is bounded by privacy.** Public reconstructability must not require total observation or retention.

4. **Embodied context is not automatically machine truth.** Spatial placement and social convention acquire computational effect only through a declared sensing and interpretation boundary.
5. **Local variation preserves a portable floor.** A local profile may vary above the shared exchange and evidence contract without silently redefining it.
6. **Dissent survives majority action.** Refusals, objections, alternatives, and unresolved claims remain inspectable without becoming the active rule.
7. **Exit and reconstruction remain possible.** A community must be able to recover its artifacts, rules, history, and active state across provider or implementation change.

2. Screen-Based Computing as Private Jurisdiction

The transcript begins by contrasting screen-based computing with communal computing. People may be physically together, but each person is inside a separate private world. Their attention, tools, data, and action surfaces are separated.

This is a constitutional problem.

A screen is not only a display. It is a private jurisdictional enclosure. The user enters a vendor-designed world, accepts that world's rules, and acts through an interface whose deeper law is hidden.

In constitutional governance terms, screen-based computing tends to produce:

- private interaction streams;
- opaque platform law;
- hidden state transitions;
- isolated attention;
- vendor-defined affordances;
- inaccessible implementation;
- weak local modification;
- limited shared inspection;
- poor civic co-presence.

Dynamicland's critique aligns with the constitutional concern that systems become ungovernable when their rules, transitions, and authority structures are hidden inside private products.

3. Communal Computing as Shared Reality

Dynamicland's alternative is communal computing: people gathered around shared physical materials, working together in the same space, seeing the same objects, and modifying the computational environment as part of the activity.

The technical and constitutional significance is that shared physical space provides a common reference frame. People can point, touch, move, annotate, negotiate, and repair together.

In constitutional governance terms:

- objects become public artifacts;
- state becomes spatially inspectable;
- rules become visible and discussable;
- collaboration becomes embodied;
- local modification becomes practical;
- computation becomes part of a shared civic field.

This strongly parallels the goal of constitutional governance: to make authority, evidence, change, and refusal visible enough that participants can reason together about what is happening.

A constitutional system should not only be correct internally. It should be legible externally.

4. Programs as Physical, Inspectable Artifacts

One of the most important technical images in the transcript is the operating system as a bulletin board, poster gallery, or binder. Programs are physical sheets of paper. People can read them, move them, edit them, gather around them, and discuss them.

This is a direct physical analogue of governed artifact architecture.

The lesson is not that all code must literally be on paper. The deeper lesson is:

> The governing structure of a system should be present in the environment where the system is used.

For UGK and MCIR, this supports the following design principle:

> Constitutional law should not be hidden in a remote codebase, invisible runtime, private platform, or undocumented convention. It should be projected as an inspectable artifact surface.

This applies to:

- profile law;
- receipts;
- refusal states;
- validators;
- authority edges;
- semantic bindings;
- conformance results;
- artifact lineage;
- publication status;

- profile succession rules.

Dynamicland's visible programs suggest a strong constitutional interface requirement: users should be able to see and interrogate the rules that govern the activity they are participating in.

5. Physical Reality as Computation Substrate

Dynamicland deliberately uses the physical world to eliminate code. A card already has position. A table already has adjacency. A binder already has ordering. A shelf already communicates storage and availability. A backpack already communicates privacy.

This is an important anti-bloat principle.

The constitutional governance version is:

> Do not encode in software what can be carried by physical reality, social practice, cryptographic identity, explicit profile law, or local custody.

This matters because governance systems can easily become over-encoded. They can try to model every human convention, every procedural nuance, and every social practice inside software. Dynamicland reminds us that some structure should remain embodied, social, local, or physical.

UGK should therefore remain lean at the substrate layer. It should not absorb all domain practice into core law. Instead:

- core law should govern admission, refusal, identity, receipts, replay, and conformance;
- local practices should be expressible in profiles;
- social norms should be preserved where appropriate;
- physical custody should count as real custody;
- privacy should sometimes be achieved by non-collection rather than encryption after capture.

This strengthens the architectural discipline of keeping MCIR low-level and stable, while allowing HLIR/profile layers to carry domain meaning.

6. Rules as Socially Negotiated Practice

The board game comparison is central. In a board game, players can change the rules because the pieces exist in the real world and the rules are socially agreed upon. In a computer game, the rules are frozen inside the developer's universe.

Dynamicland wants computation where anyone can reprogram anything at any time.

Constitutional governance should preserve the participatory spirit while adding admissibility discipline.

The constitutional form is:

> Anyone may inspect, discuss, and propose changes. Not every proposed change is automatically admitted.

The governed transition is:

proposal -> staging -> admissibility -> ratification -> receipt -> new rule state

This is the bridge between Dynamicland and UGK. Dynamicland supplies the communal rule-changing environment. UGK supplies the constitutional mechanism by which rule changes become lawful rather than arbitrary.

The key principle:

> Participatory computation requires governed mutation.

Without participation, computation becomes centrally planned. Without governance, participation becomes unbounded mutation. Constitutional governance is the discipline that allows participation without losing continuity.

7. Local Computational Culture

Dynamicland argues that computing should not be a mass-market product imposed uniformly across all communities. It should be a practice that local communities learn, teach, adapt, and maintain for their own needs and values.

This maps directly to local profile governance.

A constitutional computing substrate should allow each community to define:

- local vocabulary;
- local evidence standards;
- local workflows;
- local authority structures;
- local privacy norms;
- local publication practices;
- local refusal meanings;
- local civic priorities;
- local educational forms;
- local public interfaces.

But this local variation must not destroy interoperability. That is where the HLIR-to-MCIR structure becomes important.

A useful architecture is:

local practice -> local HLIR profile -> canonical MCIR representation -> UGK admissibility and receipt layer

In this model:

- HLIR carries local culture and domain meaning;

- MCIR preserves canonical representation;
- UGK preserves admissibility, refusal, receipts, replay, and conformance;
- GMB preserves semantic identity and governed meaning;
- publication layers preserve self-certifying address and provenance.

Dynamicland says computing should be locally grown. Constitutional governance adds that locally grown computation should still be canonically inspectable and interoperable.

8. Participatory Governance and Public Deliberation

The transcript contrasts centrally planned environments with participatory design. Built environments have concepts like democratic planning and public governance, but digital environments often lack analogous concepts. Tools like Google Docs are privately designed worlds that users inhabit but do not govern.

This is directly relevant to the constitutional-governance synthesis proposed here.

Most digital collaboration tools support group activity without supporting group sovereignty. They allow multiple people to edit, comment, and share, but the underlying rules are vendor-defined.

A constitutional public computing environment should expose:

- who may propose changes;
- who may ratify them;
- what evidence is required;
- what objections/refusals exist;
- what version is active;
- what changed since the previous version;
- what authority admitted the change;
- what receipts record the process.

This creates a pathway from collaborative editing to participatory governance.

Dynamicland's tables and walls are civic surfaces. UGK/MCIR can make those surfaces constitutionally meaningful.

9. Computational Public Space

The transcript asks a crucial question:

> What is computational public space?

This is one of the most important questions for constitutional computing.

Public infrastructure increasingly routes people into private platforms. The internet may be public infrastructure, but much of lived digital life occurs inside privately owned computational environments. The same pattern occurs in cities when public roads route people into private malls, offices, and vehicles.

Constitutional governance should distinguish:

- public protocol;
- private platform;
- shared artifact space;
- local institution;
- vendor backend;
- civic jurisdiction;
- public witness layer;
- private storage layer.

A computational public space should be:

- inspectable;
- locally governable;
- privacy-preserving;
- interoperable;
- not fully vendor-owned;
- capable of public memory;
- capable of public deliberation;
- capable of self-certifying artifact publication;
- capable of preserving dissent, refusal, and minority report;
- capable of showing the provenance of decisions.

This connects Dynamicland to the Cryptographic Receipts Provenance Layer. A public computational space needs public proof surfaces, not merely hosted documents.

10. Privacy by Non-Collection

The transcript makes a strong privacy claim: privacy is easier when the system does not collect the data in the first place.

This is important for receipt theory.

Governance does not require total capture. In fact, total capture can become surveillance. A good receipt system should not record everything indiscriminately. It should leave carefully chosen waymarks at meaningful transition points.

The constitutional privacy principle is:

> Receipt what must be reconstructable. Do not capture what can remain embodied, local, ephemeral, private, or socially governed.

This supports the crafted breadcrumb/waymark model of receipts.

A receipt should be:

- intentional;
- compact;
- placed at a meaningful transition;

- sufficient for later reconstruction;
- scoped to the governing need;
- respectful of privacy;
- not a general-purpose surveillance record.

Dynamicland's physical privacy model is instructive: public things go on the table or shelf; private things go in the backpack; sensitive things go into a locked room. Constitutional computing should preserve analogous privacy boundaries.

11. Receipts as Crafted Waymarks

The transcript implicitly supports a better metaphor for receipts than exhaust or passive logging.

Receipts are not accidental byproducts. They are deliberately placed traces that allow future reconstruction without total surveillance.

The strongest formulation is:

> Receipts are the crafted waymarks of governed action.

They are intentionally left at artfully chosen points where future reconstruction will matter. Each waymark should preserve enough information to reconstruct identity, authority, admissibility, provenance, and transition state without unnecessarily recording the whole surrounding life of the participants.

This fits Dynamicland's broader principle: computation should be visible and accountable without swallowing the whole world into data.

12. Hidden Systems Made Visible

The civic section of the transcript is perhaps the strongest alignment with ACG. It argues that democratic society cannot govern systems it cannot see.

Water, electricity, trash, internet, transit, supply chains, taxes, public service quality, and greenhouse gas emissions are all systems people depend on but generally experience only at endpoints.

The constitutional governance formulation is:

> A society cannot govern hidden systems except through superstition, proxy trust, or institutional dependence.

The proposed architecture can represent civic systems as governed flow structures:

- sources;
- transformations;
- jurisdictions;
- authorities;
- infrastructure;
- dependencies;

- receipts;
- externalities;
- costs;
- risks;
- service levels;
- outcomes.

This is a natural extension of MCIR as water engineering for governed change. Civic systems are literal and figurative flows: water, energy, waste, money, people, goods, information, emissions, authority, and responsibility.

Dynamicland asks for visibility. ACG asks for governed visibility.

13. Civic Infrastructure as CRE Domain

The transcript points toward several candidate civic-infrastructure domains:

- water systems;
- electrical grids;
- waste systems;
- transit systems;
- supply chains;
- municipal budgets;
- public service quality;
- environmental emissions;
- public health infrastructure;
- land use and zoning;
- civic deliberation processes.

Each is a system of complex change with structure, authority, state, flows, externalities, and consequences.

A civic-infrastructure governed model would represent:

- physical infrastructure;
- service flows;
- jurisdiction boundaries;
- authorities;
- maintenance actions;
- cost structures;
- public outcomes;
- environmental impacts;
- service inequities;
- evidence sources;
- dispute/refusal conditions;
- civic receipts.

The goal is not merely dashboard visualization. The goal is democratic reconstructability.

People should be able to ask:

- Where did this water come from?
- Who maintains this pipe?
- What changed after the repair?
- Which neighborhood receives which service level?
- Where did this tax dollar go?
- What supply chain did this purchase invoke?
- What emissions did this action cause?
- What authority made this decision?
- What evidence supports that claim?
- What alternatives were refused?

This is constitutional civic computing.

14. Dynamicland as Embodied Semantic Manifold

Dynamicland's computational objects acquire meaning from spatial relations:

- above;
- below;
- bottom-right;
- near;
- on the table;
- on the wall;
- in the binder;
- on the shelf;
- pointed at;
- attached to;
- part of the same activity.

This is directly relevant to Semantic Manifold Topology and GMB.

Meaning is not inside the object alone. Meaning emerges from the object's relation to a field.

In GMB terms:

- physical object = semantic node;
- spatial relation = typed edge;
- activity context = semantic field;
- program behavior = admissible continuation;
- shelf/table/backpack = custody/privacy state;
- public presentation = publication act;
- local workshop = jurisdictional context.

Dynamicland is therefore a working example of embodied semantic topology. It shows that computation can be organized around relations, contexts, and affordances rather than isolated files and screens.

This reinforces the GMB design principle:

> Meaning is structured reachability under context.

15. Public Artifact Tables and Constitutional Interfaces

Dynamicland's tables suggest a powerful interface metaphor for constitutional systems: the public artifact table.

A public artifact table is a shared space where participants can place:

- claims;
- evidence;
- receipts;
- profiles;
- objections;
- alternatives;
- maps;
- data;
- models;
- proposed rules;
- admitted rules;
- refused transitions.

The table is both physical and constitutional. It makes the state of deliberation visible.

In software terms, this could become:

- a civic review surface;
- a profile authoring surface;
- a GMB semantic editing surface;
- a governed admission and deliberation dashboard;
- an MCIR artifact inspection environment;
- a public receipt browser;
- a participatory budgeting table;
- a supply-chain trace table;
- a local policy simulation table.

The important constraint is that it must remain shared, inspectable, and locally understandable.

16. Dynamicland's Open Mutation vs Constitutional Admission

The main divergence is that Dynamicland emphasizes that anyone can reprogram anything at any time. This is appropriate for an exploratory communal computing environment, but not sufficient for high-stakes governance.

Constitutional governance requires a distinction between:

- inspection;
- proposal;
- modification;
- admission;
- execution;
- publication;
- supersession.

A participant may be allowed to inspect and propose. A profile may require stronger authority to admit or execute. Some domains may allow local rule changes by consensus. Others may require evidence, review, or external authority.

The constitutional adaptation is:

> Maximize public inspectability and proposal rights. Govern admission according to domain authority.

This preserves Dynamicland's democratic spirit without collapsing safety.

17. Universal-GSM and Corpus Placement

17.1 GMB as the governed civic meaning layer

Governed Meaning Base (GMB) is the current name in the author-confirmed lineage `CCF -> CMB -> GMB`. For communal computing, GMB can retain the meanings and standing of objects, relations, profiles, proposals, evidence, decisions, receipts, dissent, and successor states. It does not make a physical placement self-interpreting, convert local convention into universal law, or grant authority merely by storing a relation.

A universal-GSM projection should keep at least these records distinct:

- **Civic surface element:** card, sheet, model, or physical object
  Governed semantic record: artifact with source, custody, and jurisdiction
  Boundary: object identity is not semantic identity
- **Civic surface element:** spatial or social relation
  Governed semantic record: typed relation plus observer and interpretation contract
  Boundary: adjacency does not have one universal meaning
- **Civic surface element:** participant
  Governed semantic record: role, capabilities, consent, and authority scoped to a jurisdiction
  Boundary: presence is not consent or mandate
- **Civic surface element:** local rule
  Governed semantic record: profile artifact with version, standing, and amendment path
  Boundary: readable prose is not executable law

- **Civic surface element:** proposal
  Governed semantic record: candidate transition and intended effects
  Boundary: proposal is not admission
- **Civic surface element:** decision
  Governed semantic record: outcome, reasons, authority, evidence, and scope
  Boundary: majority preference is not automatically legitimate authority
- **Civic surface element:** receipt waymark
  Governed semantic record: bounded transition record
  Boundary: reconstruction evidence is not exhaustive capture
- **Civic surface element:** dissent
  Governed semantic record: preserved objection or alternative
  Boundary: retention is not adoption
- **Civic surface element:** custody/privacy state
  Governed semantic record: declared physical, social, cryptographic, or institutional boundary
  Boundary: a backpack, room, or label is not a security proof
- **Civic surface element:** successor
  Governed semantic record: revision or supersession relation
  Boundary: current standing does not erase prior state

17.2 UGK, MCIR, and portable constitutional profiles

This candidate treats UGK as a possible executable reference and receipt/certification substrate, not as the definition of communal governance. MCIR may carry canonical representation and bridge evidence, but MCIR is not authority. A portable constitutional profile should be independently meaningful, project into a local human-readable surface, lower into a declared canonical form when needed, and retain proof of what was and was not preserved.

The public artifact table is therefore a projection surface, not law by appearance. Its human view, canonical representation, active profile, and runtime effects must remain cross-checkable. No UGK, MCIR, GMB, or civic-infrastructure implementation is delivered by this paper.

17.3 Interfaces to the wider corpus

- *Constitutional Computing* [C1] supplies the wider phase shift from opaque fixed software toward inspectable and renegotiable computational institutions.
- *Sovereign Software* [C2] supplies inspectability, reconstructability, provider exit, and user sovereignty up to a chosen boundary.
- *Reasoning by Way of Receipts* [C3] supplies evidence-linked reasoning, while preserving the rule that receipts do not create truth or legitimacy.

- *The Three Hash Rule* [C4] separates source, admitted transition, and resulting artifact commitments instead of treating one digest as the whole constitutional state.
- *A Formal Theory of Work Custody* [C5] keeps physical artifact, digital artifact, logical work, version family, custody, and authorship distinct.
- *Semantic Legitimacy Before Context* [C6] requires a community to know which context may legitimately govern a decision rather than treating all nearby information as admissible.

These are architecture interfaces, not independent evidence that communal constitutional computing improves civic outcomes.

UGK

UGK provides the constitutional kernel for admission, refusal, receipts, replay, and conformance. Dynamicland shows why those mechanisms should be visible and locally usable, not hidden behind opaque automation.

MCIR

MCIR represents the canonical structure of governed change. Dynamicland shows a physical analogue: objects arranged in space whose relations determine lawful behavior.

HLIR

HLIR is the likely bridge between local human practice and canonical machine representation. Local communities should be able to author local governance in readable profile languages, then lower those profiles into MCIR.

GMB

GMB should represent meaning as relational, contextual, and embodied. Dynamicland demonstrates that object meaning emerges from placement, relation, activity, and shared practice.

Governed admission profile

A governed admission profile should distinguish proposal from admission. Dynamicland's "anyone can change anything" becomes: participants or agents may propose within declared scope; the applicable jurisdiction, authority, evidence, and profile decide what is admitted.

Cryptographic Receipts Provenance Layer

Public computational space requires public proof. A receipts provenance layer can provide self-certifying publication, witness, lineage, and replay without reducing all civic computation to a private platform.

18. Technical Requirements Suggested by the Transcript

A constitutional communal computing system should support:

1. Shared artifact visibility.
2. Local profile authoring.
3. Human-readable governance surfaces.
4. Physical or spatial metaphors for custody and publication.
5. Privacy by non-collection.
6. Intentional receipt waymarks.
7. Public deliberation workflows.
8. Self-certifying artifact addresses.
9. Profile-local vocabulary and authority.
10. Canonical lowering to portable representation.
11. Civic system visibility.
12. Participatory proposal with governed admission.
13. Public/private artifact boundary states.
14. Reconstructable decision lineage.
15. Local-first operation with optional wider publication.
16. Multi-scale artifacts: table, room, building, city.
17. Inspection before automation.
18. Refusal and objection as visible civic artifacts.

18.1 Full realization funnel

Stage	Communal-computing role
Possibility	Explore shared tables, walls, rooms, portable kits, local networks, civic maps, educational spaces, private/public boundaries, governance regimes, and beneficial, harmful, exclusionary, or incoherent variants.
Specification	Declare participants, accessibility, artifacts, relations, sensing boundary, jurisdiction, proposal rights, authority, profile law, privacy, receipts, dissent, exit, interoperability, and success/failure measures.
CM1	Enumerate admissible realizations of the specification, including costly, inconvenient, socially fragile, over-instrumented, low-accessibility, or weakly interoperable configurations.

Stage	Communal-computing role
CM2	Prefer configurations that minimize collection, expose active law, support accessibility, preserve dissent and exit, resist local capture, remain reconstructable, and outperform simpler collaboration tools on declared goals.
Instance Configuration	Freeze the exact site, participants, role assignments, artifacts, rules, sensors, local services, profile, code, state, authority, retention policy, comparison baseline, and trial duration.
Implementation Realization	Operate that configuration, mediate proposed transitions, display active rules and state, issue bounded receipt waymarks, and preserve permitted dissent and recovery artifacts.
Evidence	Measure participant comprehension, reconstructability, proposal/admission accuracy, privacy loss, accessibility, capture, refusal handling, interoperability, exit, failure recovery, and comparison with an ordinary baseline.
Revision	Preserve the prior configuration and evidence; revise the profile, interface, physical arrangement, representation, or claim when failures or better alternatives appear.

18.2 Candidate public-artifact-table pilot

A first pilot should remain smaller than the paper's civic ambition. One reversible candidate configuration is:

- four to twelve consenting participants in one local room;
- one bounded decision problem with no legal, financial, medical, employment, or safety-critical effect;
- physical proposal, evidence, authority, objection, outcome, and supersession cards;
- a local-only display of the active profile and transition state;
- no biometric collection and no audio or video recording by default;
- receipts only at proposal, admission/refusal/defer, amendment, and final-state transitions;

- an ordinary whiteboard or shared-document process as the comparison condition;
- explicit accessibility alternatives for participants unable to use the physical surface;
- export of the full local artifact and rule set into a provider-independent package at the end.

The pilot would test whether participants can identify the active rule, proposal status, authority, reasons, objections, and current state; whether an independent participant can reconstruct the transition; whether non-collected information remains absent; and whether the constitutional surface produces benefits beyond an ordinary facilitator and whiteboard. This is a proposed Instance Configuration only. It is not authorized, implemented, or evaluated by this paper.

Embodied Civic Interface Configuration Profile

Across the completed P1 corpus, Communal Computing and Constitutional Governance is the **embodied civic interface configuration profile for making public artifacts, rules, proposals, evidence, dissent, decisions, and revisions jointly inspectable in shared physical space**. It is not a replacement constitutional kernel, proof of Dynamicland behavior, or evidence that co-location automatically creates common knowledge, consent, legitimacy, or security.

- **Adjacent work:** *Universal Governance Kernel (UGK)*
  Supplies: Admission, refusal, authority, transition, and receipt primitives.
  Communal-computing boundary: A public artifact table projects selected UGK relations without becoming the kernel.

- **Adjacent work:** *Sovereign Software*
  Supplies: Inspectability, reconstructability, substitutability, fork, and provider exit.
  Communal-computing boundary: The civic profile applies sovereignty goals to communal computational space.

- **Adjacent work:** *A Formal Theory of Work Custody*
  Supplies: Identity and custody for physical, digital, and logical work.
  Communal-computing boundary: Physical possession is one custody signal, not automatic authority or semantic identity.

- **Adjacent work:** *Semantic Legitimacy Before Context*
  Supplies: Admission discipline for context allowed to govern decisions.
  Communal-computing boundary: Spatial proximity and visibility do not make an artifact admissible.

- **Adjacent work:** *Reasoning by Way of Receipts*
  Supplies: Bounded records of proposal, decision, transition, and revision.
  Communal-computing boundary: Receipt waymarks preserve reconstruction while minimizing surveillance.

A civic transition should bind:

$$C = (space, jurisdiction, participants, roles,\\ public_artifacts, proposal, authority, law, evidence,.\\ privacy, decision, dissent, effect, receipt, exit)$$

The physical surface is a projection. Position, adjacency, possession, gesture, and social convention acquire computational force only through declared sensing and interpretation. Every consequential interpretation must expose its observation boundary, alternative readings, and appeal path. Privacy by non-collection should be preferred where the governance objective does not require capture.

Participation rights, proposal rights, admission authority, execution authority, and publication standing remain separate. Majority preference is not automatically jurisdictionally sufficient. Dissent artifacts, minority reports, refusal, local fork, and provider-independent reconstruction remain visible after a decision.

The universal GSM can type
`PRESENTED_AS_PUBLIC_ARTIFACT`, `LOCATED_IN_CIVIC_SPACE`,
`INTERPRETED_UNDER_LOCAL_PROFILE`, `PROPOSED_BY_PARTICIPANT`, `ADMITTED_BY_CIVIC_AUTHORITY`,
`PRESERVES_DISSENT_ARTIFACT`, `RECEIPTED_BY_MINIMAL_WAYMARK`, `WITHHELD_BY_NON_COLLECTION`,
`FORKED_BY_COMMUNITY`,
and
`RECONSTRUCTED_AFTER_PROVIDER_EXIT`. These relations make the interface profile portable without universalizing local practice.

This integration establishes four editorial commitments: classify the work as an embodied civic interface configuration profile; adopt the civic-transition envelope; require declared interpretation and privacy-by-non-collection boundaries; and keep participation, proposal, admission, execution, dissent, publication, and exit structurally separate.

Public Artifact Correspondence and Effect Boundary

The fixed-seed comparison with Meta-Binding, GOORRT, APD, and GCML adds a public-artifact correspondence contract:

$$P = (public_artifact, representation_family, interpretation,\\ participants, authority, procedure, effect_surface, \quad .\\ observation, dissent, exit, remainder)$$

GOORRT allows public objects or models, proposed civic operations, and realized effects to remain visibly distinct. A card, table object, projection, or gesture should disclose whether it states a model, proposes a transition, records a realized state, or links among them. Spatial juxtaposition is not correspondence by itself.

GCML could express a machine-facing civic procedure after admission, but language and compiler conformance cannot establish legitimate participation, authority, consent, privacy, or physical effect. APD adds scoped containment for computational and cyber-physical effects while leaving social, distributive, reputational, and institutional harms that may be irreversible or outside the instrumented boundary.

Meta-Binding is a
`QUALIFIED_DOMAIN_PROFILE`: a shared space can test whether represented public constraints affect later behavior and whether revision authority is respected. Collective compliance, ritual, social pressure, or interface affordance remain alternative explanations and potential overbinding failures.

The universal GSM can type
`DECLARES_PUBLIC_REPRESENTATION_FAMILY`,
`PROPOSES_CIVIC_OPERATION`, `RECORDS_REALIZED_PUBLIC_STATE`,
`LOWERS_ADMITTED_CIVIC_PROCEDURE`, `CONTAINS_DECLARED_PUBLIC_EFFECT`,
`LEAVES_SOCIAL_EFFECT_REMAINDER`, and
`TESTS_CONSTRAINT_USE_IN_PUBLIC_SPACE`.

This randomized review establishes four editorial commitments: adopt the public-artifact correspondence contract; require visible object/operation/realization roles; keep GCML conformance below civic legitimacy; and require APD claims to expose uncontained social and institutional remainder.

An Under-Promoted Civic Interface Profile

The frozen P3 corpus contains no confirmed incoming dependency on this exact brief. That absence is informative: communal computing has not yet become a shared substrate elsewhere in the corpus. The paper should therefore remain a bounded civic interface and pilot profile, not a silently assumed foundation for Constitutional Computing.

$$\begin{aligned} P^* = (&place, community, participants, artifacts, shared_rules,\\ &private_zones, mutation_rights, admission,\\ &deliberation, receipts, maintenance, accessibility,\\ &safety, exit, remedy, evidence, remainder) \end{aligned}.$$

Physical visibility is not inspectability when programs, sensors, networks, models, or maintenance remain opaque. Public presence is not consent. Local participation is not representative legitimacy. Non-collection is not privacy if observation, inference, memory, or social pressure remains. Open mutation must distinguish proposal, sandbox, admission, reversible effect, emergency stop, repair, and durable public change.

A pilot should measure comprehension, participation distribution, accessibility, maintenance burden, conflict resolution, privacy, capture, physical safety,

reversibility, and continuity after facilitators or vendors leave. Dynamicland remains an inspiration and comparison subject; this paper grants it no verified conformance or deployment standing.

The universal GSM can type
`INSTANTIATES_COMMUNAL_COMPUTING_PROFILE`,
`GOVERNS_PUBLIC_ARTIFACT`, `RESERVES_PRIVATE_ZONE`, `ADMITS_COMMUNAL_MUTATION`,
`CONTESTS_PUBLIC_EFFECT`, `EXERCISES_COMMUNAL_EXIT`, and
`RETAINS_CIVIC_REMAINDER`.

This review establishes four editorial commitments: retain under-promoted profile standing; adopt the civic pilot record; separate visibility, consent, participation, legitimacy, and privacy; and require measured public governance before any broader architectural promotion.

19. Countermodels, Failure Modes, and Falsifiers

The argument must survive alternatives and harms that a shared physical metaphor can conceal:

1. **Ordinary-tool countermodel.** A facilitator, whiteboard, versioned document, and clear process may provide equal legibility at much lower cost.
2. **Performative transparency.** Rules may be visible but too technical, incomplete, stale, or disconnected from actual runtime effects.
3. **Public coercion.** A shared table can amplify conformity, status pressure, embarrassment, and retaliation.
4. **Accessibility exclusion.** Spatial, visual, tactile, linguistic, cognitive, or mobility demands can make the supposedly public surface less public.
5. **Local capture.** Local governance can be dominated by a faction, venue owner, technician, or informal hierarchy.
6. **False physical security.** A shelf, room, or backpack communicates a norm but does not prove confidentiality, integrity, or authorized access.
7. **Hidden backend control.** A communal interface can still depend on vendor services, remote policy, opaque models, or revocable accounts.
8. **Receipt surveillance.** Waymarks can become a behavioral dossier if purpose, minimization, access, retention, and deletion are not enforced.
9. **Semantic drift.** Communities can use the same portable field names while assigning incompatible meanings or authority effects.
10. **Interoperability pressure.** A canonical core can quietly erase local practice; unlimited variation can make exchange and audit impossible.
11. **Proposal flooding and denial of governance.** Broad proposal rights can overwhelm attention and prevent ordinary work.
12. **Embodied misread.** Sensing errors or ambiguous placement can assign an unintended computational meaning to a physical act.
13. **Minority erasure.** A polished final surface can hide unresolved objections and the distribution of burdens.

14. **Maintenance collapse.** A locally grown system may become unusable when its stewards leave or physical and software components age.

The proposed architecture should lose standing for a target domain when a simpler baseline matches or exceeds it on comprehension, reconstructability, agency, privacy, accessibility, and recovery; when participants cannot reliably identify active law and authority; when visible artifacts diverge from runtime effects; when non-collection claims fail; when dissent or exit is practically unavailable; or when the constitutional layer increases surveillance, exclusion, capture, or dependence. Failure in one domain does not prove all communal computing impossible, but it must constrain the scope of the claim.

20. Strategic Implication

Dynamicland strengthens the case that constitutional governance should not be framed only as safety, compliance, or control. It is also a way to make computation more public, humane, local, educational, and democratically legible.

This is strategically important because many AI governance systems will trend toward centralized surveillance: capture everything, classify everything, audit everything, and sell compliance. Dynamicland points in the opposite direction:

- collect less;
- make more visible;
- keep computation local where possible;
- use physical/social affordances;
- let communities author their tools;
- make hidden systems understandable;
- preserve public space.

Constitutional computing can become the formal discipline that tests whether this approach is sufficiently inspectable, bounded, reconstructable, accessible, and governable for more serious systems.

Shared Surfaces Need Public Governance, Not Just Visibility

Public orientation

Communal Computing and Constitutional Governance asks how embodied, locally modifiable computing could become a computational public space rather than a private platform displayed in public. Visible artifacts, local authorship, non-collection, governed admission, dissent, receipts, and portable representations can support that goal. Co-location, physicality, openness, and inspectability do not by themselves create shared understanding or legitimate authority.

Adversarial tests

1. **Visibility-intelligibility attack.** Artifacts are physically public but inaccessible through language, disability, expertise, time, culture, or interface design.
2. **Local-capture attack.** Present, powerful, technical, or property-holding participants dominate rules while absent and affected people lack voice.
3. **Physical-custody attack.** Possession or proximity is treated as secure, legitimate, durable, or privacy-preserving custody.
4. **Non-collection-shadow attack.** The primary system avoids storage while cameras, networks, logs, models, operators, or adjacent infrastructure collect equivalent data.
5. **Receipt-surveillance attack.** Reconstruction captures identities, relationships, dissent, movement, or activity beyond the minimum public purpose.
6. **Open-mutation attack.** Easy local modification bypasses admission, safety, consent, accessibility, conflict, rollback, and effect boundaries.
7. **Portable-core-colonization attack.** A shared representation or constitutional profile erases local vocabulary, practice, jurisdiction, and refusal.
8. **Public-space-analogy attack.** Software architecture claims civic legitimacy without law, maintenance institutions, funding, accessibility, remedies, and affected-community governance.

Claim and evidence ceiling

The brief offers a civic computing architecture and pilot proposal. It does not establish the accuracy of its Dynamicland descriptions, successful communal governance, common knowledge, inclusion, privacy, security, local sovereignty, portability, implementation, or public legitimacy. The author-supplied transcript remains unidentified and must be recovered and verified for a future evidence-qualified revision.

Any pilot requires affected-community co-design, accessibility, privacy and threat review, authority and amendment rules, dissent preservation, maintenance and funding, incident response, exit and export, and evaluation against existing communal and civic systems. No external research, civic process, or deployment occurred in this pass.

Reader-facing consequence

The public promise is computation people can gather around, understand, modify, contest, and carry away - with explicit protection against the familiar conversion of a shared surface into someone else's private jurisdiction.

21. Publication Standing and Evidence Limits

This paper is a conceptual and system-architecture proposal. It contains no implemented communal-computing system, validated Dynamicland integration, civic deployment, user study, interoperability proof, privacy audit, accessibility evaluation, security review, or demonstrated public-governance outcome. Its attributed Dynamicland source also remains unresolved.

22. Final Synthesis

Dynamicland asks how computation can return to shared human reality. Constitutional governance asks how that shared computational reality can remain lawful.

Together, they suggest a powerful direction:

> Computation should be local enough to belong to a community, visible enough to be understood, flexible enough to be changed, private enough to preserve human life outside data capture, and constitutional enough to preserve authority, evidence, refusal, and continuity across time.

The final thesis is:

> Communal computing provides the civic body of shared computation. Constitutional governance provides its law, memory, and admissibility structure.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Cm2.** Existing source passages use this stage only as bounded context or an obligation. The paper does not claim an independently validated contribution at this stage.

Corpus-Wide Reference Positioning

This paper argues that this technical brief asks how communal, embodied, locally modifiable computing could become a genuine computational public space rather than another private platform presented on a shared surface. Its nearest external comparison fields are governance, provenance, semantic. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds

a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763
Relationship: Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E2**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/
Relationship: Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E3**] Lawrence Lessig. *Code: Version 2.0*. Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

[**E4**] OECD. *Recommendation of the Council on Artificial Intelligence*. OECD/LEGAL/0449, adopted 2019 and updated 2024. legalinstruments.oecd.org source **Relationship:** Adjacent policy standard for accountability, traceability, transparency, safety, and human-centered values.

[**E5**] W3C. *PROV-O: The PROV Ontology*. W3C Recommendation, 30 April 2013. https://www.w3.org/TR/prov-o/ **Relationship:** Adjacent provenance standard; it models entities, activities, and agents, but does not by itself establish the paper's stronger authority or standing claims.

[**E6**] Santiago Torres-Arias et al. "in-toto: Providing Farm-to-Table Guarantees for Bits and Bytes." *28th USENIX Security Symposium*, 2019, pp. 1393-1410. usenix.org source **Relationship:** Predecessor supply-chain lineage system used for comparison; no influence claim is made unless the source head already documents one.

[**E7**] OpenSSF. *Supply-chain Levels for Software Artifacts (SLSA), Version 1.2*. 2026. https://slsa.dev/spec/v1.2/ **Relationship:** Adjacent software-supply-chain assurance standard, especially relevant to provenance and verification boundaries.

Source-head external references retained under original labels [D1] Dynamicland transcript referenced by the author source. Exact transcript artifact, speaker, title, publication date, publisher, URL, and license are unresolved. All transcript-derived descriptions and quotations remain pending primary-source verification.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Constitutional Governance*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-79353AF163A4B0F1`;
source SHA-256
`FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Theory*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D2D97C4042FD281F`;
source SHA-256
`FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF`.

Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governance Engineering.* Ableman Corpus current edition; work identity `PAPER.PUB.18749870`;
source SHA-256
`372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Universal Governance Kernel (UGK).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-1F9BFB2BF8370226`;
source SHA-256
`AA5106EA71E5DF120E141435C280DFC532715CA5DE0C75B8862BB7784ECCFACD`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Legitimacy Without Self-Ratification.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-EF2382F9397335AC`;
source SHA-256
`0186A39C808F8FDDB5C8F42384DF1A26678928434BD50A0EAE8D4947390F68A2`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Reasoning by Way of Receipts.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-04E427F6F4D94454`;
source SHA-256
`AF86A66CCD6A103DD75208500AE99148911A6453B6DFDF36A954279F21AB8489`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *The Epistemic Lineage Engine.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-E0EC3DC7EABF5060`;
source SHA-256
`CDD74D9AFAB6532645E68F7F6C9F0F7A9E472D9B8E717290522566FF8DC01F11`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Ableman Corpus references [**C1**] Adam Ableman *Constitutional Computing.* Current reviewed head:
`source-material/Constitutional_Computing_`

`Structured_Abundance.md`; SHA-256
`A1EF0D1F7B5DF84A107E986421EAA95827AEDCB801B7DA63F72BEECA6047BE8D`.

[**C2**] Adam Ableman *Sovereign Software*. Current reviewed head: `source-material/Sovereign_Software_`

`Constitutional_Computing.md`; SHA-256 `D8A2707865C6636485A564546D91F2CCF45AAC80161B94AC56A3DA5CF2EB2C13`.

[**C3**] Adam Ableman *Reasoning by Way of Receipts*. Current reviewed head: `source-material/`

`Reasoning by Way of Receipts.md`; SHA-256 `8D884B96597EA6303A324173A7CE2D8C735A952E91FBE4C9EB6509FA6D6B195F`.

[**C4**] Adam Ableman *The Three Hash Rule*. Current reviewed head: `source-material/The Three Hash Rule.md`; SHA-256 `03F60317CC9AFB36DDBCF9BF1C6D185611666CC4D9AB290BD4BCC17DF4579B7C`.

[**C5**] Adam Ableman *A Formal Theory of Work Custody*. Current reviewed head: `source-material/`

`A Formal Theory of Work Custody.md`; SHA-256 `49C38F60B95E107F8B3D864ACAFB4C49067A146C807E5248307DE7B009AD4696`.

[**C6**] Adam Ableman *Semantic Legitimacy Before Context*. Current reviewed head: `source-material/`

`Semantic Legitimacy Before Context.md`; SHA-256 `1E9058F212FCB00E42E35E927DF2E7F830BC9C9499AAFF469FE65FF8AE137B68`.

Corpus references were resolved through the verified Ableman corpus skill for internal integration. They were not treated as independent empirical evidence for Dynamicland or for civic outcomes. No external source was consulted in this pass.

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *The Three Hash Rule.* Zenodo, 2026. doi:10.5281/zenodo.20788617

Adam Ableman *Semantic Manifold Topology.* Zenodo, 2026. doi:10.5281/zenodo.20257244

Adam Ableman *A Theory of Agentic Observation.* Zenodo, 2026. doi:10.5281/zenodo.20437813

Adam Ableman *CARA: Constrained Artificial Reasoning Architecture.* Zenodo, 2025. doi:10.5281/zenodo.18099715

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Adam Ableman *Governed Semantic Compilation: Lawful Continuation, Refusal Semantics, and Epistemic Promotion in Large Language Models.* Zenodo, 2026. doi:10.5281/zenodo.18453681

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus

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DOI: 10.5281/zenodo.23847492